

**PONTIFÍCIA UNIVERSIDADE CATÓLICA DO RIO GRANDE DO
DO SUL (PUCRS)**

ISABELA ARAUJO, SANTIAGO FIRPO,
LUCAS SOARES E ISRAEL GARCIA

*Analizador de tráfego de túnel via **APIs raw socket***

Laboratório de Redes de Computadores

1. Descrição da implementação e *setup* containerizado

O projeto consiste em um monitor de tráfego em Go que captura pacotes via *raw socket* na *tun0*, interpreta *IPv4/IPv6*, reagrupa fragmentos de *IPv6* e decodifica *TCP*, *UDP* e *ICMP*. A camada de aplicação é identificada pelas portas utilizadas e uma inspeção leve de *payload*. As estatísticas globais e por cliente são mantidas em memória, com contagem de conexões, pacotes e bytes por host remoto. Uma *TUI* em terminal atualiza a cada poucos segundos, rotaciona páginas automaticamente e, opcionalmente, aplica filtros/paginadores via variáveis de ambiente (*TUI_PAGE_SIZE*, *TUI_ROTATE_SECS*, etc.). Os dados são gravados continuamente em três CSVs — internet, transporte e aplicação — já com cabeçalhos, metadados relevantes e *flush* imediato para facilitar inspeção com *tail/cat*.

Para rodar a aplicação em contêineres, foi implementado no túnel C (*traffic_tunnel*) a validação do nome de interface e tratamento fatal para falhas de *socket*, evitando comportamentos indefinidos. Também foi implementada resolução *MAC* via *ARP*, tratamento dinâmico de *IPv4/IPv6*, caminhos de erro mais robustos, e foram removidas as estruturas originais de encapsulamento *Ethernet/IP hard-coded*.

Essas mudanças permitem que o túnel opere dentro do *Docker*, encaminhando os *payloads IP* reais com os endereços mascarados *10.66.66.0/24* na *tun0* por meio do uso de *raw sockets*, modo promíscuo e a configuração *NAT*.

O *entrypoint* do servidor habilita *IP forwarding*, utiliza *NAT (MASQUERADE)*, inicia *listeners* locais em *TCP/UDP* (80, 443, 53, 123) para gerar tráfego simulado e só então lança o monitor. Já os *entrypoints* de cliente configuram seus túneis individuais, sobem *tun0*, e podem acionar um script de geração de tráfego (*generate-traffic.sh*) que emite *pings*, *HTTP(S)*, *DNS*, *NTP* e solicitações *DHCP* com *payloads* sintéticos. A emulação de tráfego pode ser desligada por meio das variáveis de ambiente.

O *Dockerfile* compila o binário Go (estágio *builder*) e o túnel C (estágio dedicado), e a imagem final *Debian-slim* inclui ferramentas de rede (*iproute2*, *iptables*, *netcat*, *dig*) e *scripts*. O *setup* inclui dois clientes e um servidor ligados por uma *bridge* no *Docker*. Todos os *containers* persistem *logs csv*.

2. Testes

O projeto foi executado por meio do *setup Docker Compose*, que inclui o monitor/servidor de tráfego, e dois clientes capazes de gerar tráfego mascarado em todos os protocolos suportados. Seguem abaixo os *logs* de execução do servidor/monitor de tráfego:

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NETWORK TRAFFIC MONITOR - Real-Time Statistics

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Uptime: 00:05:10 | Interface: tun0

GLOBAL STATISTICS

Total Packets: 3864 | Total Bytes: 368.7 KB

Network Layer:

IPv4: 3839 IPv6: 25 ICMP: 902

Transport Layer:

TCP: 2910 UDP: 52

Application Layer:

HTTP: 776 HTTPS: 299 DNS: 4

DHCP: 4 NTP: 44 Other: 1835

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PER-CLIENT STATISTICS

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Client: 10.66.66.1

Remote: 10.66.66.101

Ports: 68, 35529, 37090, 37092, 38180, ... (+4 more)

Protocols: ICMP, TCP, UDP

Seguem os logs de execução do cliente 2:

[client] Starting traffic_tunnel client with IFACE=tun0, CLIENT_UNDERLAY_IF=eth0,
CLIENT_SCRIPT=/client-config/client2.sh

[client] Launching traffic_tunnel client...

[client] Waiting for tun0 to appear... (timeout 20s)

[client] Bringing tun0 up...

[client] tun0 is ready.

[client] Generating sample traffic

[gen] Starting traffic generation...

[gen] Pinging 8.8.8.8 (3x)

PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.

64 bytes from 8.8.8.8: icmp_seq=1 ttl=1 time=91.5 ms (DUP!)

64 bytes from 8.8.8.8: icmp_seq=1 ttl=1 time=91.5 ms (DUP!)

From 10.66.66.101 icmp_seq=2 Redirect Host(New nexthop: 10.66.66.1)

From 10.66.66.101 icmp_seq=2 Redirect Host(New nexthop: 10.66.66.1)

--- 8.8.8.8 ping statistics ---

2 packets transmitted, 1 received, +371 duplicates, +2 errors, 50% packet loss, time 1001ms

rtt min/avg/max/mdev = 50.781/68.681/91.483/11.222 ms

[gen] HTTP GET (IPv4) <http://10.66.66.1/>

curl: (52) Empty reply from server

curl: (35) OpenSSL SSL_connect: SSL_ERROR_SYSCALL in connection to 10.66.66.1:443

[gen] DNS A cloudflare.com @10.66.66.1

[gen] NTP client packet to time.google.com:123 (primary)

[gen] NTP client packet to 10.66.66.1:123 (fallback)

[gen] DHCP DISCOVER to 10.66.66.1:67

[gen] Traffic generation complete.

[client] Client is running. Sleeping indefinitely.

Segue abaixo um excerto do arquivo *camada_aplicacao.csv*:

Timestamp,Protocol,Source IP,Destination IP,Protocol Info

2025-11-19 02:34:08.169,HTTP,10.66.66.102,23.192.228.80,verb=GET path=/ host=example.com
version=HTTP/1.1

2025-11-19 02:34:08.169,HTTP,10.66.66.102,23.192.228.80,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.170,HTTP,10.66.66.102,23.192.228.80,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.171,HTTP,10.66.66.102,23.192.228.80,verb=GET path=/ host=example.com
version=HTTP/1.1

2025-11-19 02:34:08.171,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.172,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.172,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.173,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.173,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.174,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.174,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.175,HTTP,23.192.228.80,172.31.200.10,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.176,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.176,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.177,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.177,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.178,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.178,HTTP,23.192.228.80,172.31.200.10,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.179,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.179,HTTP,23.192.228.80,172.31.200.10,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.180,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.180,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.181,Other,,,(amount=1)

2025-11-19 02:34:08.181,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.182,Other,,,(amount=1)

2025-11-19 02:34:08.183,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.183,Other,,,(amount=1)

2025-11-19 02:34:08.184,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.185,Other,,,(amount=1)

2025-11-19 02:34:08.185,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.186,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

2025-11-19 02:34:08.186,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.187,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.187,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.188,Other,,,(amount=1)
2025-11-19 02:34:08.188,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.189,Other,,,(amount=20)
2025-11-19 02:34:08.198,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.199,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.200,Other,,,(amount=1)
2025-11-19 02:34:08.200,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.201,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.202,Other,,,(amount=36)
2025-11-19 02:34:08.218,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.219,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.220,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.220,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.221,Other,,,(amount=36)
2025-11-19 02:34:08.238,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.239,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)
2025-11-19 02:34:08.239,HTTP,23.192.228.80,10.66.66.102,verb=UNKNOWN (no payload)

Segue um excerto do arquivo *camada_internet.csv*:

Timestamp,Protocol,Source IP,Destination IP,Protocol Number,Info,Total Bytes

2025-11-19 02:34:08.699,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=38652 flags=0x0
frag=0,40

2025-11-19 02:34:08.700,IPv4,23.192.228.80,172.31.200.10,6,ttl=63 len=40 id=27193 flags=0x0
frag=0,40

2025-11-19 02:34:08.700,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=27193 flags=0x0
frag=0,40

2025-11-19 02:34:08.701,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=46412 flags=0x0
frag=0,40

2025-11-19 02:34:08.702,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=46412 flags=0x0
frag=0,40

2025-11-19 02:34:08.703,IPv4,23.192.228.80,172.31.200.10,6,ttl=63 len=40 id=46554 flags=0x0 frag=0,40

2025-11-19 02:34:08.703,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=46554 flags=0x0 frag=0,40

2025-11-19 02:34:08.704,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=50699 flags=0x0 frag=0,40

2025-11-19 02:34:08.705,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=50699 flags=0x0 frag=0,40

2025-11-19 02:34:08.705,IPv4,23.192.228.80,172.31.200.10,6,ttl=63 len=40 id=49429 flags=0x0 frag=0,40

2025-11-19 02:34:08.706,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=49429 flags=0x0 frag=0,40

2025-11-19 02:34:08.707,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=5899 flags=0x0 frag=0,40

2025-11-19 02:34:08.707,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=5899 flags=0x0 frag=0,40

2025-11-19 02:34:08.708,IPv4,23.192.228.80,172.31.200.10,6,ttl=63 len=40 id=12407 flags=0x0 frag=0,40

2025-11-19 02:34:08.709,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=12407 flags=0x0 frag=0,40

2025-11-19 02:34:08.710,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=38652 flags=0x0 frag=0,40

2025-11-19 02:34:08.711,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=38652 flags=0x0 frag=0,40

2025-11-19 02:34:08.711,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=27193 flags=0x0 frag=0,40

2025-11-19 02:34:08.712,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=27193 flags=0x0 frag=0,40

2025-11-19 02:34:08.713,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=46554 flags=0x0 frag=0,40

2025-11-19 02:34:08.714,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=46554 flags=0x0 frag=0,40

2025-11-19 02:34:08.714,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=49429 flags=0x0 frag=0,40

2025-11-19 02:34:08.715,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=49429 flags=0x0 frag=0,40

2025-11-19 02:34:08.716,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=12407 flags=0x0 frag=0,40

2025-11-19 02:34:08.716,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=12407 flags=0x0 frag=0,40

2025-11-19 02:34:08.717,IPv4,23.192.228.80,10.66.66.102,6,ttl=57 len=52 id=63486 flags=0x0 frag=0,52

2025-11-19 02:34:08.718,IPv4,23.192.228.80,10.66.66.102,6,ttl=56 len=52 id=63486 flags=0x0 frag=0,52

2025-11-19 02:34:08.719,IPv4,10.66.66.102,23.192.228.80,6,ttl=64 len=40 id=0 flags=0x2 frag=0,40

2025-11-19 02:34:08.720,IPv4,10.66.66.102,23.192.228.80,6,ttl=64 len=40 id=0 flags=0x2 frag=0,40

2025-11-19 02:34:08.720,IPv4,10.66.66.102,23.192.228.80,6,ttl=63 len=40 id=0 flags=0x2 frag=0,40

2025-11-19 02:34:08.721,IPv4,10.66.66.102,23.192.228.80,6,ttl=63 len=40 id=0 flags=0x2 frag=0,40

2025-11-19 02:34:08.722,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=14822 flags=0x0 frag=0,40

2025-11-19 02:34:08.723,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=14822 flags=0x0 frag=0,40

2025-11-19 02:34:08.723,IPv4,23.192.228.80,172.31.200.10,6,ttl=63 len=40 id=42273 flags=0x0 frag=0,40

2025-11-19 02:34:08.724,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=42273 flags=0x0 frag=0,40

2025-11-19 02:34:08.725,IPv4,23.192.228.80,10.66.66.102,6,ttl=59 len=40 id=14822 flags=0x0 frag=0,40

2025-11-19 02:34:08.726,IPv4,23.192.228.80,10.66.66.102,6,ttl=58 len=40 id=14822 flags=0x0 frag=0,40

2025-11-19 02:34:08.726,IPv4,23.192.228.80,10.66.66.102,6,ttl=61 len=40 id=42273 flags=0x0 frag=0,40

2025-11-19 02:34:08.727,IPv4,23.192.228.80,10.66.66.102,6,ttl=60 len=40 id=42273 flags=0x0 frag=0,40

2025-11-19 02:34:08.728,IPv4,23.192.228.80,172.31.200.10,6,ttl=63 len=40 id=65407 flags=0x0 frag=0,40

2025-11-19 02:34:08.729,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=65407 flags=0x0 frag=0,40

2025-11-19 02:34:08.729,IPv4,23.192.228.80,172.31.200.10,6,ttl=63 len=40 id=36634 flags=0x0 frag=0,40

2025-11-19 02:34:08.730,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=36634 flags=0x0 frag=0,40

2025-11-19 02:34:08.731,IPv4,23.192.228.80,172.31.200.10,6,ttl=63 len=40 id=48213 flags=0x0 frag=0,40

2025-11-19 02:34:08.732,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=48213 flags=0x0 frag=0,40

2025-11-19 02:34:08.732,IPv4,23.192.228.80,172.31.200.10,6,ttl=63 len=40 id=33593 flags=0x0 frag=0,40

2025-11-19 02:34:08.733,IPv4,23.192.228.80,10.66.66.102,6,ttl=62 len=40 id=33593 flags=0x0 frag=0,40
2025-11-19 02:34:08.734,IPv4,23.192.228.80,10.66.66.102,6,ttl=55 len=52 id=63486 flags=0x0 frag=0,52
2025-11-19 02:34:08.735,IPv4,23.192.228.80,10.66.66.102,6,ttl=54 len=52 id=63486 flags=0x0 frag=0,52
2025-11-19 02:34:08.736,IPv4,10.66.66.102,23.192.228.80,6,ttl=64 len=40 id=0 flags=0x2 frag=0,40
2025-11-19 02:34:08.736,IPv4,10.66.66.102,23.192.228.80,6,ttl=63 len=40 id=0 flags=0x2 frag=0,40

Segue um excerto do arquivo *camada_transporte.csv*:

Timestamp,Protocol,Flags,Source IP,Source Port,Destination IP,Destination Port,Total Bytes

2025-11-19 02:33:27.273,TCP,ACK,10.66.66.1,80,10.66.66.101,45108,64
2025-11-19 02:33:27.274,TCP,ACK,10.66.66.101,45108,10.66.66.1,80,52
2025-11-19 02:33:27.275,TCP,SYN ACK,10.66.66.1,80,10.66.66.101,45108,60
2025-11-19 02:33:27.277,TCP,ACK,10.66.66.1,80,10.66.66.101,45108,52
2025-11-19 02:33:27.277,TCP,ACK,10.66.66.1,80,10.66.66.101,45108,64
2025-11-19 02:33:27.285,TCP,SYN,10.66.66.101,45112,10.66.66.1,80,60
2025-11-19 02:33:27.286,TCP,SYN ACK,10.66.66.1,80,10.66.66.101,45112,60
2025-11-19 02:33:27.287,TCP,SYN,10.66.66.101,45112,10.66.66.1,80,60
2025-11-19 02:33:27.288,TCP,SYN ACK,10.66.66.1,80,10.66.66.101,45112,60
2025-11-19 02:33:27.291,TCP,SYN ACK,10.66.66.1,80,10.66.66.101,45112,60
2025-11-19 02:33:27.292,TCP,ACK,10.66.66.101,45112,10.66.66.1,80,52
2025-11-19 02:33:27.292,TCP,ACK,10.66.66.101,45112,10.66.66.1,80,52
2025-11-19 02:33:27.293,TCP,ACK,10.66.66.101,45112,10.66.66.1,80,52
2025-11-19 02:33:27.294,TCP,PSH ACK,10.66.66.101,45112,10.66.66.1,80,130
2025-11-19 02:33:27.294,TCP,ACK,10.66.66.1,80,10.66.66.101,45112,52
2025-11-19 02:33:27.295,TCP,ACK,10.66.66.101,45112,10.66.66.1,80,52
2025-11-19 02:33:27.296,TCP,ACK,10.66.66.1,80,10.66.66.101,45112,52
2025-11-19 02:33:27.296,TCP,SYN ACK,10.66.66.1,80,10.66.66.101,45112,60
2025-11-19 02:33:27.297,TCP,PSH ACK,10.66.66.101,45112,10.66.66.1,80,130
2025-11-19 02:33:27.298,TCP,ACK,10.66.66.1,80,10.66.66.101,45112,64
2025-11-19 02:33:27.300,TCP,ACK,10.66.66.1,80,10.66.66.101,45112,52

2025-11-19 02:33:27.301,TCP,ACK,10.66.66.1,80,10.66.66.101,45112,52
2025-11-19 02:33:27.302,TCP,ACK,10.66.66.1,80,10.66.66.101,45112,64
2025-11-19 02:33:28.204,TCP,RST ACK,10.66.66.1,80,10.66.66.101,45108,52
2025-11-19 02:33:28.206,TCP,RST ACK,10.66.66.1,80,10.66.66.101,45112,52
2025-11-19 02:33:28.207,TCP,RST ACK,10.66.66.1,80,10.66.66.101,45108,52
2025-11-19 02:33:28.208,TCP,RST ACK,10.66.66.1,80,10.66.66.101,45112,52
2025-11-19 02:33:28.209,TCP,SYN,10.66.66.101,54602,10.66.66.1,443,60
2025-11-19 02:33:28.210,TCP,SYN ACK,10.66.66.1,443,10.66.66.101,54602,60
2025-11-19 02:33:28.211,TCP,SYN,10.66.66.101,54602,10.66.66.1,443,60
2025-11-19 02:33:28.212,TCP,SYN ACK,10.66.66.1,443,10.66.66.101,54602,60
2025-11-19 02:33:28.213,TCP,SYN ACK,10.66.66.1,443,10.66.66.101,54602,60
2025-11-19 02:33:28.213,TCP,ACK,10.66.66.101,54602,10.66.66.1,443,52
2025-11-19 02:33:28.214,TCP,FIN ACK,10.66.66.101,54602,10.66.66.1,443,52
2025-11-19 02:33:28.215,TCP,ACK,10.66.66.101,54602,10.66.66.1,443,52
2025-11-19 02:33:28.215,TCP,ACK,10.66.66.1,443,10.66.66.101,54602,52
2025-11-19 02:33:28.217,TCP,ACK,10.66.66.101,54602,10.66.66.1,443,52
2025-11-19 02:33:28.218,TCP,SYN ACK,10.66.66.1,443,10.66.66.101,54602,60
2025-11-19 02:33:28.219,TCP,FIN ACK,10.66.66.101,54602,10.66.66.1,443,52
2025-11-19 02:33:28.220,TCP,ACK,10.66.66.1,443,10.66.66.101,54602,64
2025-11-19 02:33:28.221,TCP,ACK,10.66.66.101,54602,10.66.66.1,443,52
2025-11-19 02:33:28.222,TCP,ACK,10.66.66.1,443,10.66.66.101,54602,52
2025-11-19 02:33:28.222,TCP,ACK,10.66.66.1,443,10.66.66.101,54602,64
2025-11-19 02:33:28.223,TCP,RST ACK,10.66.66.1,443,10.66.66.101,54602,52
2025-11-19 02:33:28.224,UDP,,10.66.66.102,32927,216.239.35.4,123,76
2025-11-19 02:33:28.225,UDP,,10.66.66.102,32927,216.239.35.4,123,76
2025-11-19 02:33:28.226,TCP,RST ACK,10.66.66.1,443,10.66.66.101,54602,52
2025-11-19 02:33:28.228,UDP,,10.66.66.101,39480,10.66.66.1,53,80
2025-11-19 02:33:28.229,UDP,,10.66.66.101,39480,10.66.66.1,53,80
2025-11-19 02:33:28.263,UDP,,10.66.66.101,43493,200.192.232.8,123,76
2025-11-19 02:33:28.264,UDP,,10.66.66.101,43493,200.192.232.8,123,76

3. Conclusão

O projeto demonstra um monitor de tráfego em tempo real robusto para ambientes de túnel/proxy, integrando captura em raw sockets, classificação multi-camada e visualização via TUI com histórico em CSV. Além das implementações de *parsing* de pacotes nos níveis de aplicação, transporte e rede, o projeto também demonstra as modificações necessárias para executar um experimento de rede local física em um ambiente com containers. Entre as modificações, a principal foi a resolução manual de endereços MAC por meio do protocolo ARP.